TASK-3:

INPUT:

```
#include <stdio.h>

enum { LIGHT = 0, FAN = 1, AC = 2, HEATER = 3, PUMP = 4 };

unsigned char switch_on(unsigned char reg, int dev)
{
   return reg | (1u << dev);
}

unsigned char switch_off(unsigned char reg, int dev)
{
   return reg & ~(1u << dev);
}

unsigned char toggle(unsigned char reg, int dev)
{
   return reg ^ (1u << dev);
}

int is_on(unsigned char reg, int dev)
{
   return (reg & (1u << dev)) != 0;
}

void show(unsigned char reg)
{
   for (int bit = 7; bit >= 0; bit--)
      putchar(((reg >> bit) & 1u) ? '1' : '0');

   putchar('\n');
}

int main(void)
{
   unsigned char reg = 0;

   reg = switch_on(reg, LIGHT);
   reg = switch_on(reg, AC);

   show(reg);

   reg = toggle(reg, FAN);
   reg = switch_off(reg, LIGHT);

   show(reg);
```

```
    printf("AC is %s, PUMP is %s\n",
           is_on(reg, AC) ? "ON" : "OFF",
           is_on(reg, PUMP) ? "ON" : "OFF");

    return 0;
}
```

OUTPUT

